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Lab: Lab 4

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a. Program Description:

The purpose of this program is to calculate interest rate and mortgage based on user inputs. The program uses functions defined in different programs which are called from a main program which is where the program execution begins.

There are in total 4 functions defined.- Mortgage, Interest calculator, print user and main. Main function is the starting point of program and based on inputs different functions are called.

b. Program Source Code:

```
import InterestCalculator
import Mortgage
from printuser import *
print_user()
def main():
    selections()
    while(True):
        try:
            command = int(input("Select one of the command number above "))
            if command == 1:
                InterestCalculator.interestCalculator()
                selections()
            elif command == 2:
                Mortgage.mortgage()
                selections()
            elif command == 99:
                print("Have a nice day...")
                exit()
            else:
                print("Error:Command not recognised")
        except Exception as err:
            print("Error:Command not recognised")
def selections():
    msg1 = "Calculate simple interest"
    msg2 = "Calculate Mortgage Payment"
    msg3 = "Quit the program"
    length = len(msg2) + 10
    print("-" * length)
    print("{:8}".format("1"), msg1)
    print("{:8}".format("2"), msg2)
    print("{:8}".format("99"), msg3)
    print("-" * length)
if __name__ == "__main__":
    main()
```

c. Program Output

```
C:\Users\shwetachauhan\AppData\Local\Programs\Python\Python310\python.exe C:/Users/shwetachauhan/PycharmProjects/Course_scripts/userInput.py
CNET-142 - Shweta Chauhan, Lab 4 - Menu Function
2022-04-10 23:21:33.651365
-----
1      Calculate simple interest
2      Calculate Mortgage Payment
99     Quit the program
-----
Select one of the command number above 10
Error:Command not recognised
Select one of the command number above 1
Enter the starting principle, <= 0 to quit: 1000
Enter the annual interest rate: 1.0
How many times per year is the interest compounded? 100
For how many years will the account earn interest? 1
At the end of 1.0 years you will have $ 1,035.62 with interest earned $ 35.62

Enter the starting principle, <= 0 to quit: 0
Exiting Simple Interest program ...
-----
1      Calculate simple interest
2      Calculate Mortgage Payment
99     Quit the program
-----
Select one of the command number above 2
Enter the loan amount, 0 to quit: 150000
Enter the loan interest rate % : 2.85
Enter the loan term (number of years): 30
For the loan amount of $150,000.00 for 30 years with interest rate 2.85 %
The monthly payment is $620.34
```

```
-----
Exiting Simple Interest program ...
-----
1      Calculate simple interest
2      Calculate Mortgage Payment
99     Quit the program
-----
Select one of the command number above 2
Enter the loan amount, 0 to quit: 150000
Enter the loan interest rate % : 2.85
Enter the loan term (number of years): 30
For the loan amount of $150,000.00 for 30 years with interest rate 2.85 %
The monthly payment is $620.34
Total amount paid for this loan is $223,320.99
Total Interest paid for this loan is $73,320.99
Enter the loan amount, 0 to quit: 200000
Enter the loan interest rate % : 2.5
Enter the loan term (number of years): 15
For the loan amount of $200,000.00 for 15 years with interest rate 2.50 %
The monthly payment is $1,333.58
Total amount paid for this loan is $240,044.12
Total Interest paid for this loan is $40,044.12
Enter the loan amount, 0 to quit: 0
Exiting Mortgage Program ...
-----
1      Calculate simple interest
2      Calculate Mortgage Payment
99     Quit the program
-----
Select one of the command number above 99
Have a nice day...
```